PREP 3, LESSONS 5–6

The point that does not fit

Best-fit lines; anomalous results; conclusions; directly proportional

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Describe the effect of a systematic error on a set of readings. DN 861 [1]

It shifts every reading in the same direction by the same amount.

2 State which variable is plotted on the x-axis of a graph. DN 885 [1]

The independent variable

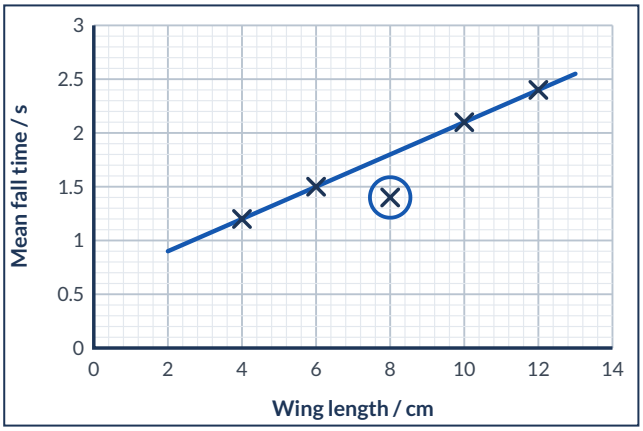
3 State two things every axis of a graph must have. DN 887 [1]

A label (the quantity) and a unit

Key ideas LESSONS 5–6

Show your working

4 A student drops paper helicopters with different wing lengths and times how long each takes to fall. The graph shows the mean fall times.



a Draw a ring around the anomalous result. **(8 cm, 1.4 s) ✓** [1]

b Draw a best-fit line. **One ruled straight line through the other four points, ignoring 8 cm ✓** [1]

c What should the student do about the anomalous result? [1]

Check it and repeat the 8 cm drops; if it is still odd, leave it out of the best-fit line ✓

d Is fall time directly proportional to wing length? [1]

yes (no) Explain.

The line does not pass through the origin (0, 0) ✓

5 Write a conclusion for this investigation. Use values from the graph. [2]

As the wing length increases, the fall time increases ✓

e.g. from 1.2 s at 4 cm to 2.4 s at 12 cm (values with units) ✓

6 The student timed each drop with a stopwatch. Name one random error and how to reduce it. [1]

Reaction time starting or stopping the watch; take repeats and calculate a mean ✓

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